

ROS 2 Clean-Slate Session and One-Shot Script Guide

When working with ROS 2, lingering background processes or DDS shared memory files can persist even after closing all terminals. These leftovers can interfere with new launches, causing unreachable nodes, communication errors, or unexpected behavior. Starting every session from a “clean slate” ensures that all components — daemons, nodes, DDS instances, and network ports — are reset.

This process is especially important when developing or testing hardware-integrated systems (like UR5e robots, Jetson devices, or embedded ROS 2 nodes), because any stale connections or mismatched domain IDs can block the new ROS 2 graph from initializing properly.

Step-by-Step: Creating the One-Shot Script

1. Open a terminal and create a script file using a heredoc. This allows writing a whole script at once:

```
cat > ~/ros2_ses_clean.sh <<'EOF'
#!/usr/bin/env bash
set -e
source /opt/ros/humble/setup.bash || true
ros2 daemon stop || true
pkill -f "ros2|python.*launch|rviz2|gzserver|gzclient|ign|rosbridge|micro_ros" 2>/dev/null || true
sleep 1
sudo rm -rf /dev/shm/fastdds_* /dev/shm/iceoryx_* 2>/dev/null || true
rm -rf ~/.ros/log/latest
for p in 30001 30002 30003 30004 50002 11311; do
  sudo ss -lptn "sport = :$p" | awk 'NR>1 {print $7}' | sed 's/pid=//;s/.*//' | xargs -r sudo kill -9
done
ros2 daemon start || true
echo "Clean slate. Now re-source your workspace and launch."
EOF
```

2. Make it executable:

```
chmod +x ~/ros2_ses_clean.sh
```

3. Run it whenever ROS 2 behaves unexpectedly:

```
~/ros2_ses_clean.sh
```

4. What each command does:

- **source /opt/ros/humble/setup.bash** — ensures ROS 2 environment variables are active.
- **ros2 daemon stop/start** — resets the ROS 2 discovery daemon, clearing cached graph info.
- **pkill -f ...** — kills any remaining ROS 2 or Gazebo-related processes.
- **rm -rf /dev/shm/...** — removes stale DDS shared memory segments used by FastDDS or CycloneDDS.
- **rm -rf ~/.ros/log/** — clears logs that can confuse relaunching.
- **for p in ...** — kills processes blocking critical ROS/UR ports.
- **echo ...** — gives feedback that the environment is clean.

5. Optional Enhancements:

Move the script to `*/usr/local/bin*` to make it globally accessible: `sudo mv ~/ros2_ses_clean.sh /usr/local/bin/ros2-ses-clean` Then you can simply type `**ros2-ses-clean**` to reset your ROS 2 session.

Summary

This one-shot script ensures a consistent, interference-free ROS 2 environment. By cleaning daemons, DDS, and network artifacts, you guarantee reproducible behavior each session, especially useful when testing robot drivers, DDS configurations, or multi-machine setups.